

Albert Einstein

The "father of modern PHYSICS"

Einstein is the world's most famous scientist. His theories changed forever the way we look at the Universe.

Early years

Albert Einstein, pictured here with his younger sister, Maja, was born in 1879 in Ulm, Germany. After school, he worked as a clerk in a **patent office** in Switzerland, checking applications for electrical devices. In his spare time, he developed radical theories about **LIGHT AND TIME**. In 1905, he published five scientific papers that would *change the world*.



His most famous equation showed that mass (m) and energy (E) are interchangeable. Even a tiny piece of matter (such as a pea) contains HUGE amounts of energy locked within its atoms. The energy in the object is the same as its mass times the speed of light (c), squared.

$$E = mc^2$$



Time warp

In his theory of **SPECIAL RELATIVITY**, Einstein showed the Universe is a pretty weird place. He suggested that **space and time are linked** and that they are flexible and can change, depending on who is looking at them. He explained that ***the faster you travel, the slower time passes for you***, and that light has a speed limit of 186,000 miles (300,000 km) per second.

He paved the way for...



The first **NUCLEAR BOMB** went off in a massive explosion in 1945. It proved Einstein's $E=mc^2$ equation, and released the energy locked in atoms.



Einstein developed a theory of **LASERS**. These are used to read CDs, DVDs, and **BARCODES**... and are often seen in science fiction films!